

# [D.3.1] Universal AI: Solomonoff Induction

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Iliad Intensive

June 2026

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- Constructed as a single **Bayesian mixture**  $\xi$  over a class  $\mathcal{M}$  of candidate environments, weighted by a prior  $w_\nu$ .
  - Solomonoff:  $\xi$  (eventually) predicts as well as the truth  $\mu$ .
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- Choose  $\mathcal{M}$  = all computable environments, prior  $w_\nu = 2^{-K(\nu)}$  ( $K$  is the “complexity” of the environment). Assumption “ $\mu \in \mathcal{M}$ ” becomes “*the universe is computable*” and the prior  $w_\nu = 2^{-K(\nu)}$  encodes *Occam's razor* (simpler universes are more likely).

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Per-step squared prediction error  $s_t^\mu$

$$s_t^\mu = \sum_{x_{<t}} \mu(x_{<t}) \left[ \sum_{x_t} (P(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t}))^2 \right]$$

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- **Goal:** build a  $P$  that makes cumulative error  $S_\infty^\mu$  *small*.
- **Idea:** don't bet on one model. Hedge your bets and mix over a whole class  $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$  of potential environments.

# The Bayesian mixture $\xi$

## Definition

Class  $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$  with prior weights  $w_\nu > 0$ ,  $\sum_\nu w_\nu = 1$ .

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- **Key fact (mixture is a predictor):** its one-step prediction is a *posterior-weighted* average of the members' predictions (you prove this today!)

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- **Posterior update is multiplicative:**  $w(\nu \mid x_{1:t}) = w(\nu \mid x_{<t}) \cdot \frac{\nu(x_t \mid x_{<t})}{\xi(x_t \mid x_{<t})}$ :  
we reweight by how well  $\nu$  predicted vs.  $\xi$  (Bayes' rule!)

# Main result: the cumulative prediction bound

## Cumulative bound

$$S_{\infty}^{\mu} := \sum_{t=1}^{\infty} \sum_{x_{<t}} \mu(x_{<t}) \sum_{x_t \in \mathbb{B}} (P(x_t | x_{<t}) - \mu(x_t | x_{<t}))^2 \leq -\ln w_{\mu}$$

- Total squared error over *all time* is finite, bounded by the prior penalty on  $\mu$ .
- Higher prior on the truth implies smaller total error.
- There are infinitely many environments. Cannot choose a uniform prior.  
What to choose?

# Specializing to the Solomonoff prior

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- **Interpretation:** the total prediction error is (roughly) bounded by the **complexity** of the true environment. Simpler worlds are learned faster.
- This is the formal version of **Occam's razor**.

# Other properties of the Solomonoff prior

- **Bounded mistakes:** on a deterministic sequence, predicting the most likely bit  $\hat{x}_t = \arg \max_{x \in \mathbb{B}} \xi(x \mid x_{<t})$  makes at most  $-2 \ln w_\mu$  wrong guesses *ever*.

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- **Pareto optimality:** no other predictor beats  $\xi$  on *every*  $\nu \in \mathcal{M}$  simultaneously (for KL loss *and* squared loss).
- **Misspecification** ( $\mu \notin \mathcal{M}$ ): the bound degrades gracefully,

$$S_n^\mu \leq -\ln w_{\hat{\mu}} + D_n(\mu \parallel \hat{\mu})$$

where  $\hat{\mu}$  is the closest model in  $\mathcal{M}$  to  $\mu$ .

# Why care? Applications, approximations, critiques

- **A yardstick for prediction.** Solomonoff's bound [Sol78] is tight up to additive  $\sqrt{\cdot}$  terms [Hut01] and Pareto-optimal for general losses [Hut03]; it dissolves classic puzzles of Bayesian confirmation (zero-prior, black ravens) [Hut07] and grounds a Bayesian theory of induction [RH11].
- **Computable cousins that work.** Shrink  $\mathcal{M}$  and the same mixture becomes an algorithm: context tree weighting [WST95] is the exact Bayes mixture over all variable-order Markov models, in linear time; MDL [Grü07] replaces the mixture by its best single code; the speed prior [Sch02] trades universality for computability.
- **Prediction is compression.** The Hutter Prize scores intelligence by compressed size of Wikipedia; LLMs turn out to be strong general-purpose compressors [DRD<sup>+</sup>24].
- **Deep learning through this lens.** Meta-training nets on UTM-generated data amortises Solomonoff induction [GMGH<sup>+</sup>24]; transformers conjectured as its best extant approximation [YW25];  $K$ -complexity explains the simplicity bias behind generalisation [GFRW24, MRVPL25].
- **The critiques.** No elegant computable universal predictor exists [Leg06]; different UTMs give non-equivalent priors [WSH13]; the “Occam's razor” reading is contested: the real assumption is computability of the world [Ste16, Ste26].

# Takeaways

- **Solomonoff** solves induction: mix over all computable hypotheses, weighted by simplicity. Total prediction error  $\leq K(\mu) \ln 2$ .

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- **AIXI** lifts this to action: the Bayes-optimal policy over the same universal mixture. It learns to predict *and* to act as well as if it knew the world.
- **The catch:** both are *incomputable* ( $K$  is incomputable;  $\mathcal{M}$  is intractable to enumerate). They are gold standards, not computable algorithms, and the prior's  $2^{-K(\nu)}$  hides a dependency on the choice of universal Turing machine.

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